

LAINIO, Sweden

WMO: 020860

Lat: 67.761N

Lon: 22.352E

Elev: 317

StdP: 97.57

Time Zone: 1.00 (EUC)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-28.6	-33.4	0.2	-30.9	-30.8	0.2	-28.5	11.0	-2.2	9.7	-2.6	1.1	320	0.774

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	22.2	14.5	20.0	13.4	18.1	12.5	15.4	20.0	14.4	18.4	13.3	16.8	3.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	10.1	16.7	12.6	9.4	15.6	11.5	8.8	14.7	44.2	19.7	41.2	18.6	38.4	16.7	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	7.8	6.7	-34.1	25.0	4.1	1.4	-37.1	26.1	-39.5	26.9	-41.8	27.7	-44.8	28.7		
			-34.2	16.8	4.1	1.6	-37.2	17.9	-39.6	18.9	-41.9	19.8	-44.9	20.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.1	-13.5	-11.2	-7.3	-2.5	4.1	10.7	12.6	10.3	5.1	-1.5	-8.5	-11.7		
	DBStd	10.59	8.27	7.14	4.83	4.58	3.55	3.90	2.95	3.34	3.25	5.41	6.21	8.20		
	HDD10.0	4238	728	593	538	375	187	37	8	37	150	357	556	673		
	HDD18.3	7081	986	827	796	625	441	229	180	248	396	615	806	931		
	CDD10.0	202	0	0	0	0	5	58	89	47	3	0	0	0		
	CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0		
	CDH23.3	22	0	0	0	0	0	6	14	2	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	2.8	2.4	2.4	2.8	3.0	3.2	3.4	3.2	2.8	2.6	2.9	2.3	2.4		
(15) Precipitation	PrecAvg	550	34	28	26	28	42	65	86	68	50	47	42	36		
	PrecMax	724	60	54	75	77	78	114	157	112	124	92	85	94		
	PrecMin	391	6	5	5	9	10	13	30	21	11	20	13	6		
	PrecStd	88	13	14	16	17	20	29	35	25	30	20	19	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	3.0	4.6	9.2	20.2	24.5	25.4	23.6	15.9	10.0	2.9	3.0		
		MCWB	0.9	0.8	1.4	4.3	11.5	15.2	17.6	15.7	11.4	6.2	1.2	1.0		
	2%	DB	1.4	1.0	2.2	6.8	15.8	22.1	22.0	20.0	13.9	8.1	1.5	1.3		
		MCWB	-0.4	-0.4	-0.3	2.8	9.0	14.4	14.6	14.0	10.3	5.8	0.5	-0.2		
	5%	DB	-0.4	-0.4	0.7	5.2	12.5	19.7	20.0	17.6	12.0	6.7	0.6	0.0		
		MCWB	-1.7	-1.3	-0.8	2.1	7.1	12.7	13.7	12.9	9.2	4.8	-0.2	-0.9		
	10%	DB	-2.1	-1.8	-0.6	3.7	10.0	17.4	18.1	15.8	10.1	5.2	-0.2	-1.3		
		MCWB	-3.0	-2.5	-1.8	1.1	5.6	11.2	12.9	12.0	7.8	3.7	-0.7	-2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	1.2	1.5	5.0	12.0	16.6	18.4	16.3	13.1	7.6	1.8	1.4		
		MCDB	2.3	2.4	3.7	7.3	20.0	21.7	23.1	21.8	14.3	8.3	2.2	2.6		
	2%	WB	-0.4	-0.2	0.2	3.4	9.6	15.0	15.6	14.9	11.3	6.3	0.7	0.2		
		MCDB	1.1	0.6	1.5	6.0	15.2	20.4	19.7	18.3	13.2	7.5	1.2	1.0		
	5%	WB	-1.5	-1.2	-0.7	2.3	7.4	13.4	14.4	13.9	9.4	4.8	0.0	-0.8		
		MCDB	-0.3	-0.4	0.5	4.6	11.9	19.0	18.7	16.6	11.3	6.4	0.5	-0.1		
	10%	WB	-3.1	-2.5	-1.7	1.5	5.9	11.9	13.5	12.7	8.2	3.8	-0.8	-2.1		
		MCDB	-2.2	-1.8	-0.6	3.4	9.5	16.2	17.0	15.0	9.6	5.3	-0.1	-1.3		
(35) Mean Daily Temperature Range	MDBR		7.5	7.7	7.9	7.9	7.3	7.5	7.5	6.9	6.4	5.7	6.5	7.8		
	5% DB	MCDBR	6.5	6.8	8.4	8.8	12.1	11.4	10.3	9.4	8.8	6.7	4.9	7.8		
		MCWBR	5.9	6.0	6.8	6.0	6.9	6.0	5.0	4.9	5.3	5.1	4.5	7.2		
	5% WB	MCDBR	6.4	6.4	6.9	7.3	11.7	10.5	8.8	7.6	7.2	5.9	4.6	7.5		
		MCWBR	5.9	5.7	5.8	5.2	6.9	6.1	4.7	4.4	4.6	4.8	4.3	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.180	0.242	0.278	0.308	0.327	0.333	0.339	0.324	0.294	0.263	0.346	N/A		
	taud		1.927	2.114	2.210	2.240	2.445	2.492	2.487	2.511	2.569	2.353	2.796	N/A		
	Ebn at Noon		136	573	765	842	870	872	853	829	775	565	184	0		
	Edh at Noon		10	47	77	100	92	91	88	77	57	37	13	0		
(44) All-Sky Solar Radiation	RadAvg		0.04	0.53	2.04	3.98	4.87	5.03	4.71	3.41	1.84	0.62	0.10	0.00		
	RadStd		0.00	0.06	0.12	0.17	0.47	0.35	0.46	0.25	0.23	0.10	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page